Applied Machine Learning - Assignment (Q3, Q4, Q5 Detailed)

Question 3: Naive Bayes Classification

Given dataset:

Email	Offer	Money	Class
E1	Yes	Yes	Spam
E2	Yes	No	Spam
E3	No	Yes	Spam
E4	Yes	No	Not Spam
E5	No	No	Not Spam

Step 1: Prior Probabilities

P(Spam) = 3/5 = 0.6

P(Not Spam) = 2/5 = 0.4

Step 2: Conditional Probabilities

Probability	Value
P(Offer | Spam)	2/3 = 0.67
P(Money | Spam)	2/3 = 0.67
P(Offer | Not Spam)	1/2 = 0.5
P(Money | Not Spam)	0

Step 3: Naive Bayes Calculation

Class	Calculation	Result
Spam	0.6 × 0.67 × 0.67	0.269
Not Spam	0.4 × 0.5 × 0	0

Step 4: Final Decision

Since Spam probability is higher, the email is classified as Spam.

Diagram (Concept Flow)

Features (Offer, Money) → Apply Probabilities → Compute Scores → Compare → Predict Class

Question 4: KNN

Point	Distance
A(1,1)	2.24
B(2,2)	1
C(3,3)	1
D(6,5)	4.24

Nearest neighbors: B, C, A

Prediction: Red

Question 5: Text Processing

Word	Doc1	Doc2
machine	1	1
learning	1	1
data	1	0
mining	1	0
models	0	1